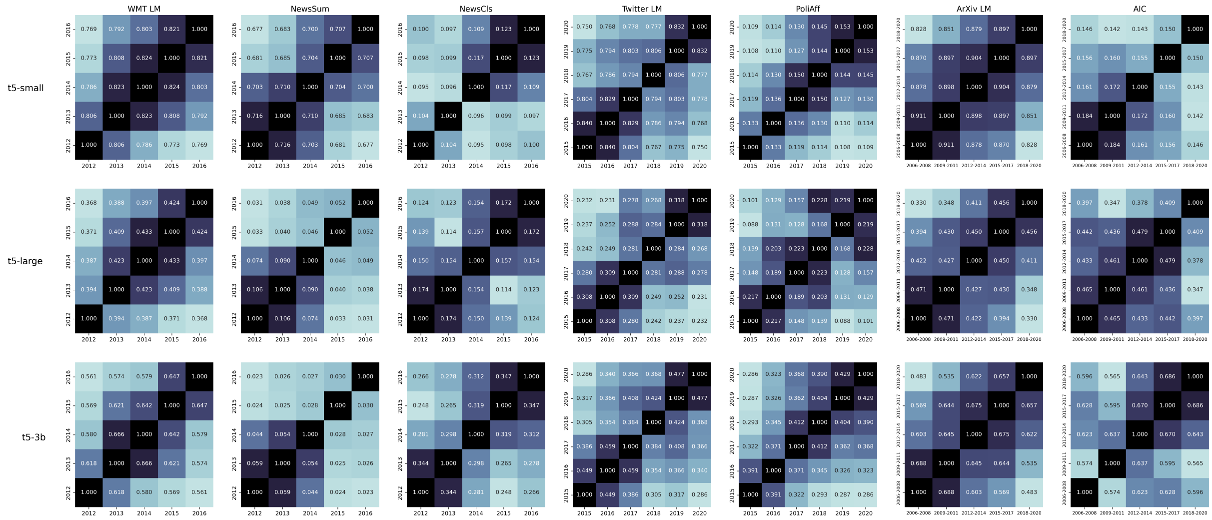

Figure 10: Cosine similarities between all pairs of year time vectors for all tasks and model sizes.

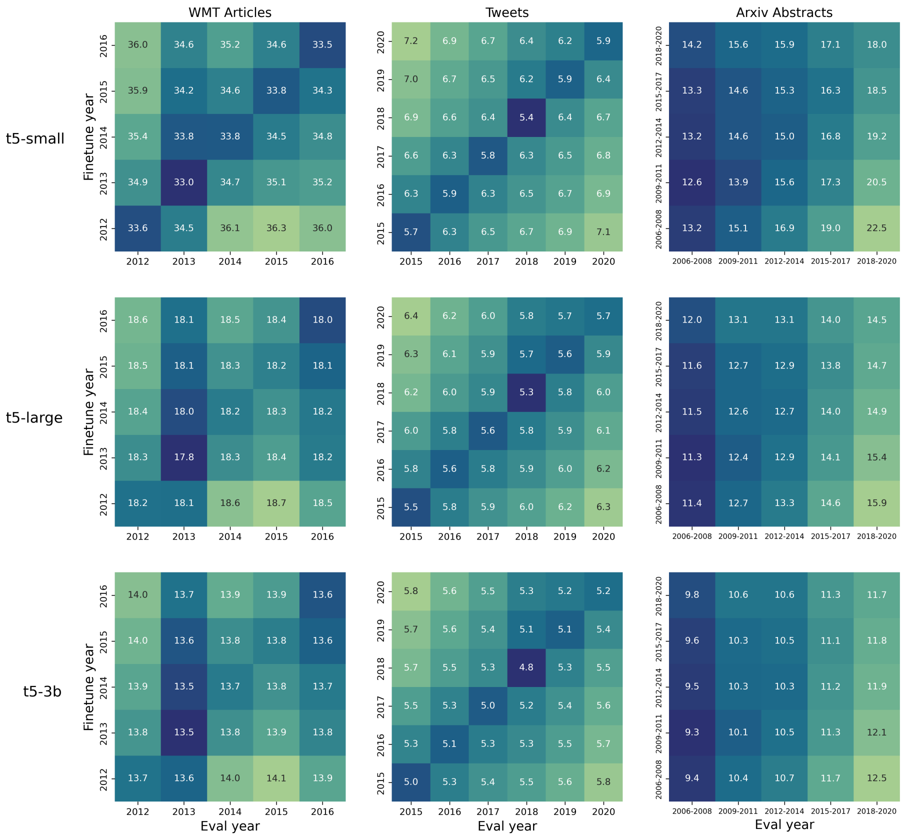

Figure 11: Yearly language modeling perplexity decay on three tasks and three T5 sizes.

Swapped Params	WMT LM	NewsSum	NewsCls	Twitter LM	PoliAff	ArXiv LM	AIC
None	*35.72*	*35.11*	*0.7232*	*6.69*	*0.5903*	*18.18*	*0.8224*
Feed Forward	35.31	35.17	0.8162	**13.25**	0.6174	18.21	0.8500
Attention	36.23	34.49	0.7986	14.95	0.6095	19.24	0.8644
Embeddings	36.13	34.30	0.7232	16.65	0.5902	19.29	0.8192
Non-Embedding	**34.57**	**37.24**	**0.8760**	**13.46**	**0.7991**	**17.37**	**0.8845**
All	*33.51*	*38.89*	*0.8759*	*5.79*	*0.7999*	*15.75*	*0.8845*

Table 5: **We can improve performance on a target time by swapping out weights with a time vector finetuned on that time.** T5-small start-year finetuned model performance on the end-year split for each task (e.g. finetuning on 2015 for PoliAff and evaluating on 2020). We compare the baseline start-year model (none swapped) to versions with various parameter weights from the target-year model, and the target-year model itself (all swapped).

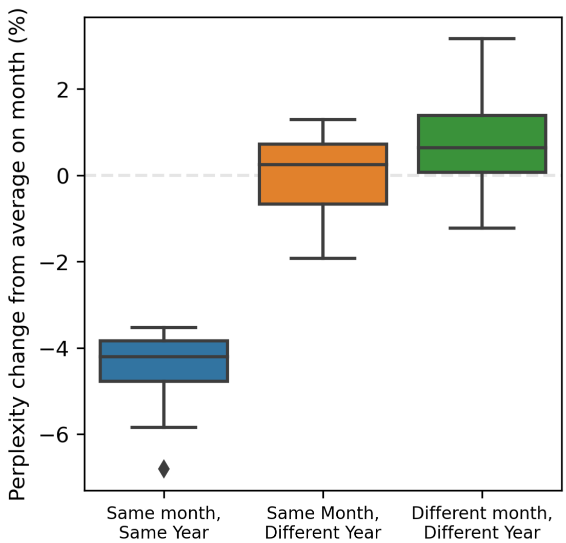

Figure 12: **Seasonality makes a small, but noticeable impact on monthly misalignment.** Distribution of perplexity change from the mean for aligned finetuning and evaluation months (left, mean=-4.36), seasonal "stripes" (middle, mean=0.04), and all finetuning and evaluation combinations which share neither the same month nor year (right, mean=0.77).

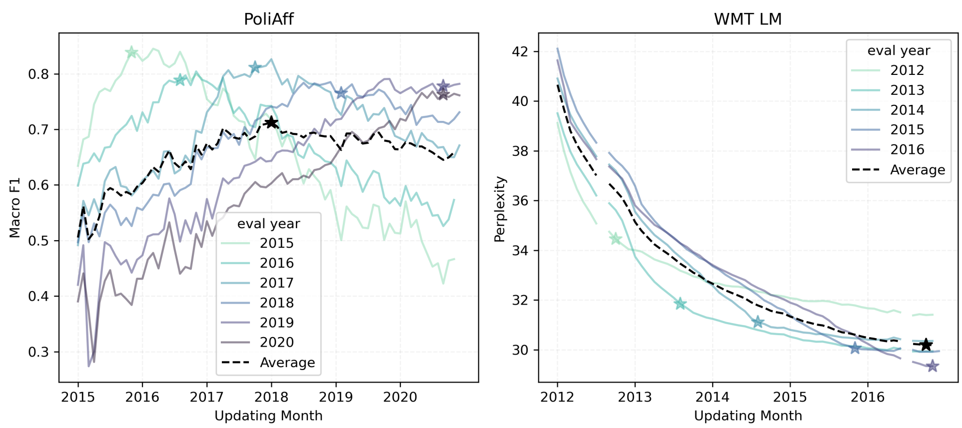

Figure 13: **In online settings, language model performance degrades on earlier time periods.** We show macro F1 and perplexity on each year split of PoliAff and WMT LM respectively after sequentially finetuning T5-small on each new month of task data. PoliAff performance over all years plateaus after finetuning on months up to 2018. WMT performance continues to improve with more data, but perplexity decrease slows on earlier years. Starred points are where performance on a year is best relative to the average performance on all years.

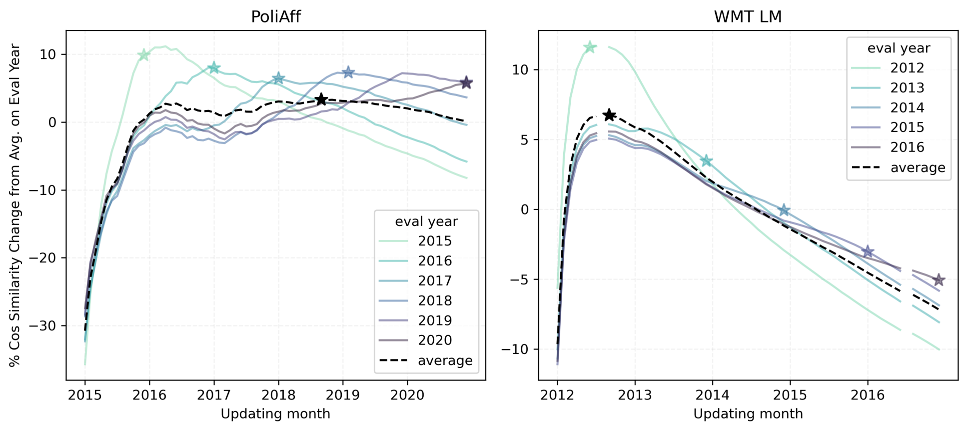

Figure 14: **Cosine similarity between an online time vector and a year vector peaks relative to other years after updating on data for that year.** We show cosine similarity between each monthly checkpoint of online T5-small time vectors and yearly vectors for PoliAff and WMT LM. To account for overall decreases in similarity as online time vectors are updated, we normalize similarities to each year vector by the mean similarity to that year over all checkpoints. We star the point for each year vector where its cosine similarity to the online model is largest relative to the average on all years.